

Toward Changing Users behavior with Emotion-based Adaptive Systems

Mina Alipour
mial@mmmi.sdu.dk

University of Southern Denmark
Odense, Denmark

Karthik Vaidhyanathan
karthik.vaidhyanathan@iiit.ac.in
IIIT Hyderabad
Hyderabad, India

Mahyar T. Moghaddam
mtmo@mmmi.sdu.dk

University of Southern Denmark
Odense, Denmark

Mikkel Baun Kjærgaard
mbkj@mmmi.sdu.dk
University of Southern Denmark
Odense, Denmark

ABSTRACT

Interactive computer systems' designers emphasize the importance of considering humans, their emotions, and behaviors as first-class entities. Emotions are integral parts of human nature, and ignoring that can lead the interactive systems to failure, low quality, or discomfort. User interfaces (UIs) are increasingly becoming adaptive to users' various characteristics, intending to improve users' satisfaction, performance, and decisions. However, the previous approaches proposed for supervising such adaptations are not effectively adopted in real-life problems. This paper proposes the novel approach to adapting UIs to users' emotions using Model-Free Reinforcement Learning (MFRL). The approach aims to maximize applying the essential adaptations and minimize the unnecessary ones towards users' task completion and satisfaction. We chose emergency evacuation training as a suitable evaluation domain since people experience intense emotions in potential danger. We performed experiments with a mobile application we developed that acts as a recommender system in emergency training. By taking contextual input of the users' basic emotions from face recognition, the application intelligently adapts its UI to quickly lead people to safe areas while arousing target emotions. The research includes literature analysis, surveys, and further adopting an iterative process in implementation and experimentation. The evaluation process confirms the efficiency and effectiveness of the MFRL in iterations, as well as compared to other possible UI adaptation techniques, i.e., rule-based and sequential adaptation.

KEYWORDS

Emotions, User Interface, Self-adaptation, Reinforcement Learning, Emergency.

ACM Reference Format:

Mina Alipour, Mahyar T. Moghaddam, Karthik Vaidhyanathan, and Mikkel Baun Kjærgaard. 2023. Toward Changing Users behavior with Emotion-based Adaptive Systems. In *UMAP '23: Proceedings of the 31st ACM Conference on User Modeling, Adaptation and Personalization (UMAP '23)*, June 26–29, 2023, Limassol, Cyprus. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3565472.3595614>

1 INTRODUCTION

Are current interactive computer systems able to intelligently adapt to dynamic environmental situations [1, 38]? Can those systems observe and react to what I, as a human, want to see on my screen [47]? Should I communicate my preferences manually every time I feel like seeing something [54]? ***Can my smile adapt my interactive system?***

Emotions can be used to inform the adaptation of interactive systems and their user interfaces (UIs). Emotions are a primary part of the human essence, leading to specific behaviors and impacting well-being. Thus, understanding and considering emotions in real-time interaction is crucial for leading the users to specific behaviors [11, 36]. Some researchers [10, 46] introduced UI self-adaptation approaches, such as model-based [49], rule-based [3], or optimization-based [28]. The problem is that human behavior can hardly be modeled, optimized, or structured within the rules. Humans may show fluctuating, unpredictable and irrational behaviors due to their individual, social, or environmental context. Lack of consideration for those human-related dynamics can lead the interactive systems to failure, low quality, or discomfort.

To fill this gap, we propose a Model-Free Reinforcement Learning (MFRL) approach for UI adaptations to emotions. MFRL is a trial-and-error approach where the agent has to learn by taking actions from states repeatedly, which is known best for dynamic human behavior [45]. The type of MFRL used in this paper is called Q-learning [27]. This type of MFRL can be used when the state space and the action space are finite, meaning all possible states and actions are known. Q-learning maps all possible actions to all possible states in a Q-table, known as state-action pairs. The agent will take actions that result in changes to the state. The new state is checked against a reward table when an action is taken. The state maps to either a positive or a negative value in the reward table. The agent uses the reward/punishment value to calculate a value for taking that specific action from that specific state, aka. a value for that state-action pair. The value for the state-action pair

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

UMAP '23, June 26–29, 2023, Limassol, Cyprus

© 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-1-4503-9932-6/23/06...\$15.00
<https://doi.org/10.1145/3565472.3595614>

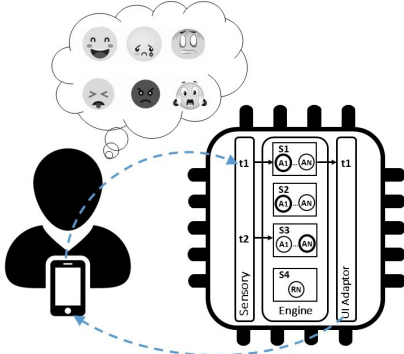


Figure 1: MFRL for UI adaptation to emotions.

is stored in Q-table as a value called the Q-value. Through trial and error, the agent will keep updating the Q-values in the Q-table. In a high-quality Q-table, the table's Q-values represent a good or even the best action to take from a particular state.

Figure 1 shows how we adopt the MFRL approach as a supervisor for emotion-based UI adaptation. The approach is built around the *MAPE-K* control loop [22, 51], which includes *monitoring* raw emotion data, *inferring* and *analyzing* emotion values, *planning* and decision-making for adaptation to *execute*, with a global *knowledge-base*. In our approach, the knowledge base that feeds the other run-time MAPE-K elements receives input from design-time activities, i.e., emotion models, temporal aspects, design parameters, and baseline Q-Tables. As shown in the figure, the built-in camera on a device detects and infers six types of Ekman basic emotions (namely joy, sadness, surprise, fear, disgust, and anger) [13] plus neutral (*one of the state attributes in MFRL*) and sends it to MFRL adaptation engine. The engine receives the emotional states each time and assigns reward/punishment values. Based on the values, a specific UI adaptation occurs (*an action in MFRL*) among various UI configurations (*the other state attribute in MFRL*) to possibly reduce the emotions associated with punishment and improve the emotions associated with reward. Q-Table values correspond to selecting an interface (*action*) from the set of interfaces (*state*).

As a **case study**, we apply the approach to a mobile application called EvacuationApp, developed for emergency evacuation training, to teach people how to safely and quickly reach a safe zone in emergencies. This app is a suitable case since people tend to show intensive emotions in simulated danger [2, 21], and being a training application, it does not impose a real risk to human safety. We first surveyed 50 participants to collect insights and then implemented and evaluated the EvacuationApp in two iterations. We first conducted evaluations with 24 participants performing four experiments each, and second, with 27 participants performing six experiments each, resulting in 258 experiments in total. In EvacuationApp, citizens place themselves in a situation of hazard and need to follow an evacuation route [4–6, 35] to reach a safe zone. The users' location and their facial emotion data are continuously tracked. The location information is used to give contextual awareness to people about their run-time position and surrounding environment. The emotion data is used to change the UI to impact their emotions (and consequently their mobility behavior) to improve the users' quality of experience. Our main objective was to prove *i)* the effectiveness of MFRL compared with other adaptation

approaches, such as rule-based and sequential, *ii)* the efficiency of MFRL in iteration, and *iii)* the improvement that MFRL causes in the task (evacuation) completion and users' satisfaction.

The main contributions this paper makes are: *i)* A survey performed to understand the emotional consequences of UIs that could be re-used for similar research; *ii)* The novel MFRL-based adaptation approach and its adaptation engine developed for the community to be used in all interactive systems with users' emotions as a primary concern (as long as there is a mechanism to capture the emotion and trigger an adaptation); *iii)* Application of the approach on a real case.

The remainder of this paper is organized as follows. Section 2 presents a literature review. Section 3 presents the approach and how it is built at both design-time and run-time. The research method is detailed in Section 4. Section 5 discusses the implementation of various pieces of the emergency training application and its adaptation engines, and Section 6 reports the results of conducted experiments. Section 7 presents the lessons learned, and Section 8 explains the threats to validity. Section 9 concludes the paper and outlines future work directions.

2 STATE-OF-THE-ART REVIEW

User interface adaptation to emotions is interdisciplinary and requires unifying social, cognitive, and computer sciences. Firmenich et al. [15] discuss the two main techniques for *UI adaptations* in HCI that are known as tailor-made and rule-based adaptations. Tailor-made systems focus on defining adaptation at design-time or the beginning of a user session so that users can directly select or configure the desired version. Rule-based adaptation is instead based on programming the rules at design-time and updating a part or the whole UI at run-time based on dynamic contextual events. For instance, Zheng et al. [55] proposed a rule-based approach to adapt a user interface to various contexts, such as the user's age, color preferences, and weather status. Märtin et al. [30] developed a UI adaptation environment to improve the user's satisfaction while interacting with a system. Galindo et al. [16] developed a rule-based UI adaptation engine that can be integral to any web technology.

Apart from the methods mentioned above, machine learning methods [33] such as *reinforcement learning (RL)* could be adopted as an adaptation supervisor. Reinforcement learning focuses on learning what to do and how to map situations into actions to maximize a numerical reward signal. It divides the problem into four components: a policy, a reward function, a value function, and an internal environment model. There exist two main approaches in reinforcement learning: model-based and model-free. Model-based methods rely on planning as their primary component, while model-free methods primarily rely on learning. In a recent paper, Todi et al. [49] present an adaptive UI approach using model-based reinforcement learning to improve user satisfaction and avoid surprises. The approach was applied to an adaptive menu to enhance the menu organization for the user. However, the difficulty is that in the model-based approach, a model for each individual should be created. The agent would generate acceptable results in adapting to only the human for which the model is created. Yechiam et al. [53], Maia et al. [29], and Gläscher et al. [18] state that MFRL and its model-free approach of using Q-learning allow enacting individual

level fine grained adaptations; however, it is not yet discovered in UI adaptation to the emotion domain.

The UI adaptation techniques mentioned above could be ideally triggered by users' detected **emotions** [1]. Kleinginna et al. [24] define emotions as reactions to events tied up with a person's physiological, affective, behavioral, and cognitive state. For instance, fear is an emotional reaction to a (negative) threatening situation, and joy is a reaction to (positive) goal fulfillment. Brosch et al. [9] emphasize that emotions could be driven by external events called stimuli, like nice weather or a dark-themed UI. As emotions involve changes in our biological system, they can be detected from various measurements of biological changes [17] (e.g., heart rate or muscles' electrical changes) but also by the impacts of these changes on exposed behaviors, for instance, by measuring changes in the face [12], gesture [19], or voice [44]. This research considers facial expressions as the source of emotions. Based on Konar et al. [25], facial expression analysis is the most common modality of emotion recognition.

There are two main theoretical categories in implying facial emotions: *the categorical models* [14] and *dimensional models* [41]. Categorical theories define emotions using tags from identified collections of emotions. The most prominent approach in this category is Ekman's basic (universal) emotions [13]. It represents six basic emotions of *anger*, *disgust*, *fear*, *joy*, *sadness* and *surprise*, which cover positive and negative emotions. Based on Savran et al. [43], and Meudt et al. [32], such emotions are defined as universal facial expressions and are generally stable over races, cultures and ages and are consistent even in people blind by birth. The dimensional model expresses emotions in a two or more-dimensional form. With the Circumplex Model of Affect theory, a theory in the dimensional category, Penser et al. [41] suggest that emotions have at least two dimensions in a circular space. These two dimensions are valence, which shows pleasure or displeasure (feeling good or bad), and arousal, which defines activation or deactivation (intensity of emotion) [52]. Bradley et al. [8] discuss other dimensional instruments (e.g., Self-Assessment Manikin, SAM) and theories (e.g., Pleasure-Arousal-Dominance, PAD), which consist of valence, arousal, and dominance space. We adopt the Ekman basic emotions model since it is universal, well-established, and stable over various categories of users.

This literature review studied UI adaptation, RL as adaptation supervisor, and emotions' importance in UI adaptation. Unifying and harmonizing these three areas are crucial for understanding User interface adaptation to emotions. Apart from understanding the gaps, we take advantage of the knowledge we gathered about emotion models, MFRL, and alternative adaptation approaches to establish design-time and run-time aspects concretely.

3 APPROACH

3.1 Design-time

The design-time step includes the activities performed before the implementation to guarantee suitable functionality and high quality. As shown in Figure 2, the design-time layer investigates which emotional models should be adopted, what are the time-related aspects of emotions, how the UI should be designed, and how we could support our decision-maker with some insights. In Section 2, we mentioned **adopting emotion models** for classifying emotions.

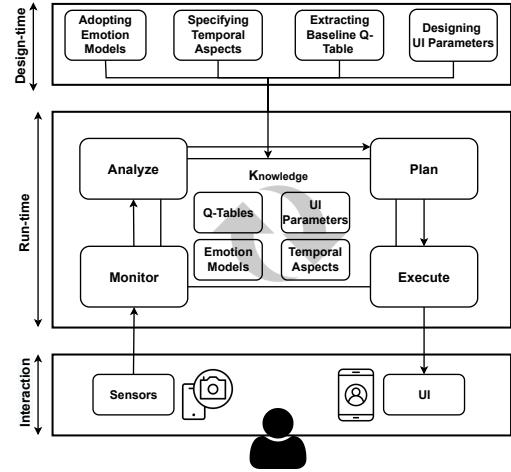


Figure 2: The emotion-driven adaptation approach.

Emotion models play a key role in our approach since they impact the interpretation of emotions based on what humans demonstrate, mainly by their faces. We mentioned that among dimensional and categorical models, we choose Ekman's basic emotions [14] under categorical models since we must specify not only the valence but also the exact emotions at each time. Ekman is known as a universal emotion model that is not dependent to age, gender, and race [32, 43]. Though there is a debate on considering such an emotion model as universal, it still is a very well-performed emotion model for facial expression recognition. In addition to models, there are **temporal aspects** of emotions to consider. This dimension emphasizes the application of UI adaptation in a proper frequency or at the right time. Since we are adopting the Q-learning approach, we should overcome the time issue where the agent has to explore all possible actions from all possible states. Thus, we performed a survey to **create that first Q-Table**. Sections 4 and 6 explain this process in detail. As shown in Figure 2, the last design-time component is **designing UI parameters**. In fact, the UI parameters are calculated variables expressing the adaptation that must be made to the UI. For instance, detecting confusion in the user's emotions would lead to increased guidance with a pop-up. In such a case, a UI parameter could have various states, such as visible and hidden. In line with the temporal aspect mentioned above, UI parameters can be associated with an additional character called the maximum adaptation. In the EvacuationApp case study, the UI parameters are font size, background color, pop-up, and map type, and they could be changed from one state to another.

3.2 Interaction

Self-adaptive systems constantly interact with their surrounding environment [51]. They receive sensory inputs and give adaptation commands to the part of the environment they control. The interaction layer is a part of the environment and includes **sensors**, physical or software components and the screen's surface where the **interface** is rendered. In our approach, all sensors provide raw data at run-time that are transferred to the run-time layer. The interface is used for specific goals (such as routing, gaming, etc.) and is supposed to adapt continuously to the detected contextual situation. For example, in the *EvacuationApp* case study, the central

sensor in facial emotion recognition is the built-in mobile camera, and the mobile screen facilitates the rendering of the interface. The participant's facial expression is captured continuously through video feeds using their mobile phone camera. At run-time, these video feeds are sent to *FaceAPI*, an emotion analysis library, at a rate of ten frames per second (video is a collection of image frames), which infers emotion from face muscles' situation and movement data.

3.3 Run-time

The run-time layer handles the adaptations when the software system is running. As previously explained, the process to follow is conceptualized using MAPE-K control loop [51] that includes four main steps: *monitor*, *analyze*, *plan*, and *execute* under a comprehensive *knowledge* base [39]. Sensors produce raw data, in our case, emotions from the user's face. In such a process, the **monitor** step filters raw data into insightful data and watches sensors changes and infers values' changes. Afterward, the **analyze** computes inferred values from raw values. In **plan** step, the adaptation process performs the decision-making by planning the application of the UI adaptation and getting input from the control loop's knowledge base. The control loop's **knowledge** base contains the historical data and the inputs from design-time step's actions (explained above) [34]. Following the plan step, and if the adaptation is not stopped, the **execute** step modifies the interface. In EvacuationApp, the camera feeds data into the FaceAPI library, which uses pre-trained models to predict which emotions are likely to be expressed. FaceAPI then passes that information to the adaptation engine. The engine then plans which adaptation(s) should be triggered. When a decision has been made, that info is passed to the adaptation context within execute, which alters its UI state.

The process described above highly depends on the adaptation decision-maker, which is the core of the plan step. The role of the **decision maker** is to select the optimal action (change of UI) by taking into account the current emotional state of the user as well as the current UI of the application. To this end, our approach uses an RL technique, Q-learning [50]. The key part of using Q-learning is to divide the problem into a set of states, S , actions, A , and rewards, R . The state space of our Q-learning approach is determined by two important attributes: i) Emotion categories, E and ii) Interface configurations, I .

Definition 3.1. (Emotional Categories) We define Emotional Categories, E as a discrete set $\{e_1, e_2, \dots, e_n\}$ where e_j represents the emotion of the user captured by the user application by mapping the analysis of the facial image obtained to a unique label, $I \in \mathcal{L}$.

Based on this, the state space, $S \subseteq E \times I$ is defined over the set of possible emotional categories and the set of all interface configurations in I .

For instance, let $E = \{joy, surprise, fear, disgust, angry, sad, neutral\}$ represent the emotional categories and let $I = \{dark, light, 3dmap\}$ represent a set of interface configurations. Then the state space S consist of all possible combinations of $E \times I$ such that $S = \{joy_dark, joy_3dmap, \dots\}$.

Action space, A , consists of a set of actions $\{a_1, a_2, a_3, \dots, a_m\}$ such that $a_j \in A$ denotes the selection of a configuration s_j in I . Rewards, R on the other hand is a set $\{r_1, r_2, \dots, r_n\}$ where $r_j \in \mathbb{Z}$ denotes the reward value for attaining a state, $s_j \in S$.

Algorithm 1 Interface Configuration Selection Algorithm

Require: :

- 1: States $S = \{s_1, s_2, s_3, \dots, s_n\}$
- 2: Actions $A = \{0, 1, 2, \dots, m\}$ ▷ represents the selection of each $i \in I$
- 3: Labels $L = \{l_1, l_2, l_3, \dots, l_p\}$ ▷ emotional categories
- 4: Rewards $R = \{r_1, r_2, \dots, r_n\}$ ▷ Reward for each of the state
- 5: **procedure** DECISION-MAKER($S, A, L, R, I, \alpha, \gamma$) ▷ Find the state of the system from the emotional category and current interface configuration
- 6: $E \leftarrow \text{identify_category}(I, L)$ ▷ Get Category from Forecasts
- 7: $s \leftarrow (E, i)$ ▷ combine category and interface configuration to form the state
- 8: $r \leftarrow R[s]$ ▷ Reward for attaining the state, s
- 9: $(s', a') \leftarrow \text{argmax}_a Q(s, a)$
- 10: $Q'(s, a) = (1 - \alpha) * Q(s, a) + \alpha * (r + \gamma * \max(Q(s', a)))$
- 11: $a \leftarrow a'$ ▷ The action to reach that state
- 12: **return** a

S and A together form a $n \times m$ matrix where n denotes the number of states and m denotes the number of actions. Q-Table forms the heart of the Q-learning approach where each value corresponds to a (s, a) pair and its value denotes the relevance of selecting an action, a from a state s . In other words, it denotes the benefit of selecting an interface configuration a_j given the current interface configuration in use and the user's emotion. The selection process leverages this property to select the most appropriate interface configuration (considering the given user emotion and the present UI configuration) during the process of UI adaptation. The complete algorithm based on Q-learning is presented in algorithm 1. It uses two key parameters, α and γ where $0 < \alpha \leq 1$ denotes the learning rate, which represents the importance given to the learned observation at each step, and $0 < \gamma \leq 1$ denotes the discount factor, which can be considered as the weight given to the next action (selection of an interface configuration).

The algorithm first identifies the emotional category from the facial image of the user, E based on the labels, L . The E identified along with the running interface configuration, I is combined to identify the current state, s , inside the Q-Table (lines 6-7). The algorithm then assigns a reward for attaining the state, s . Following this, the maximum value of the state-action pair, (s', a') , that can be reached from the current state is identified (lines 8-9). The Q-table is then updated using the q-function (line 10), and the action (instance, $\hat{s}$) is returned (lines 11-12). In this manner, for every query, q , our algorithm selects the most appropriate interface configuration i by selecting the action that maximizes the reward. By assigning negative rewards (penalties) to a selection of interface configuration that leads to an undesirable emotional category such as anger, the approach ensures that any incorrect selection is penalized in the form of a high negative reward, and this means as the time progresses, the algorithm continuously improves the selection process to ensure that the UI is adapted to maintain an emotional balance of the user by ignoring configurations that can lead to high penalties.

4 RESEARCH METHOD

The main objective of this research is to explore and further develop a UI adaptation approach that improves users' task completion and their target emotional state. Figure 3 summarizes the methodology [26] we followed to establish the concept. We followed three interlinked steps to achieve our objective. The results of each step feed into the starting point of each subsequent step. The study was backed by our **own experience** as human-computer interaction researchers, human behavior modelers, and software engineers,

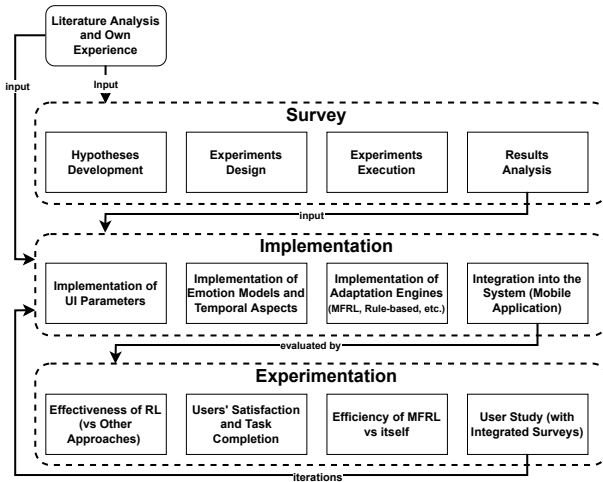


Figure 3: The research methodology.

where we collaborated with industrial practitioners and governmental decision-makers. We also gained insights from state-of-the-art studies.

In Figure 3, the second layer is the **survey**. In line with guidelines [31, 48], we distributed an online survey to our audience and gained insights into design, development, and emotions' consideration in EvacuationApp. We followed a structured question-and-answer format with multiple-choice and explanatory questions to allow the participants to discuss their opinions and ideas. As shown in Figure 3, our survey activity consists of *hypothesis development*, *experiments' design*, *experiment execution*, and *results' analysis*. In *hypotheses development* step, we develop our hypotheses based on the knowledge gathered from the literature. We formulated our hypotheses by following the SMART mnemonic [23]. The SMART acronym stands for Specific, Measurable, Attainable, Relevant, and Testable.

- **H1.** participants' joy intensity level is triggered by changes in the font size as a UI parameter.
- **H2.** participants' surprise intensity level is triggered by showing a pop-up dialogue and/or Satellite v.s. Street map view.
- **H3.** participants' fear intensity level is triggered by the background color as a UI parameter.

In the *Experiments' Design* step, we informed the participants about the high-level topic of the survey without giving them any hint to avoid any bias. We designed and deployed an online requirement analysis survey to ground our system design and development. The survey aims to understand the users' requirements for using an application to learn about evacuations in a potential hazard. We investigated: 1. What type of information would the participants like to see in the application for learning evacuation (i) what features are needed, e.g., routing, news sharing, voice guidance, hazard state, finding relatives, first-aid information, civil protection information; ii) how could these features be visualized for instance with large letters or options on-screen or with medium icons) 2. What emotions do participants experience during the learning sessions (in their previous experience on preparedness sessions, i) what are the points they find (not) useful; ii) what emotions and how intense

have they experienced, if any) 3. What emotions are involved in various types of adaptation in evacuation learning (what emotions and how intense do they feel if they see a mobile interface adaptation from picture 1 to picture 2, i.e., before and after an adaptation)?

In *experiments execution* step, the experiment was taken using Qualtrics [7], a web-based online survey platform. The experiment duration was approximately 20 minutes. We allocated an attention check question [40] to ensure the validity of the survey responses. The majority of questions were designed in an open-ended style to gain insights. We targeted the time allocation of around 20 minutes length. The survey was conducted remotely. We received 50 valid responses. 51% of the participants were female, and the remaining 48% were male. In the *results analysis* step, the authors discussed the survey results analysis from Qualtrics and reflected them into the design and development of EvacuationApp. After analysis, we compared and tuned the literature review and survey results. This process resulted in a better understanding of user expectations and potential emotions using training apps for an emergency. Therefore the survey supported the subsequent implementation and experimentation steps.

In **Implementation and Experimentation** layers, we went through an iterative process as it is expected that the app could improve and some requirements may change during the development. The process model is inspired by *Software Engineering: A Practitioner's Approach* [42], which is described as the recommended approach. It is intended that the stakeholders will be included early and often in the development process. The implementation and experimentation are thoroughly explained in Sections 5 and 6.

5 IMPLEMENTATION

We created the *EvacuationApp* as a *training* mobile application that leads people to safe areas in case of disaster. The routing is shown based on real-time contextual information such as individuals' positions, risk situation, and crowdedness. *EvacuationApp* is developed as a React progressive web application in *TypeScript*. The backend is a *NodeJS* application interacting with a *MongoDB* database. The backend is also written in *TypeScript*. Our React application has integrated several pieces: *maps*, *emotion detection software*, and *adaptation engines*. We used the *FaceAPI* library to infer emotions from the user's face every 100ms to be added to the adaptation engines.

In addition to MFRL, we employed other approaches, such as sequential and rule-based adaptation, to compare. As shown in Figure 4, we incorporated four types of adaptations to training users for a safe evacuation, namely i) *Pop-up (Hidden-Visible)*, ii) *FontSize (Medium-Large)*, iii) *MapType (SatelliteView-StreetView)* iv) *BackgroundColor (Light-Dark)*.

5.0.1 Sequential Engine. The first approach performs predefined random adaptations to the UI every twenty seconds. We chose twenty-second intervals because emotions take, on average, twenty seconds to reach the highest intensity due to external stimuli [20]. The adaptation sequence is chosen randomly among the adaptation mentioned above.

5.0.2 Rule-based Engine. In rule-based [1], the adaptation occurs based on predefined rules as the logic in an adaptation engine within a control loop. Each rule has a property that demonstrates its priority compared to other rules. The goal is to arouse three target

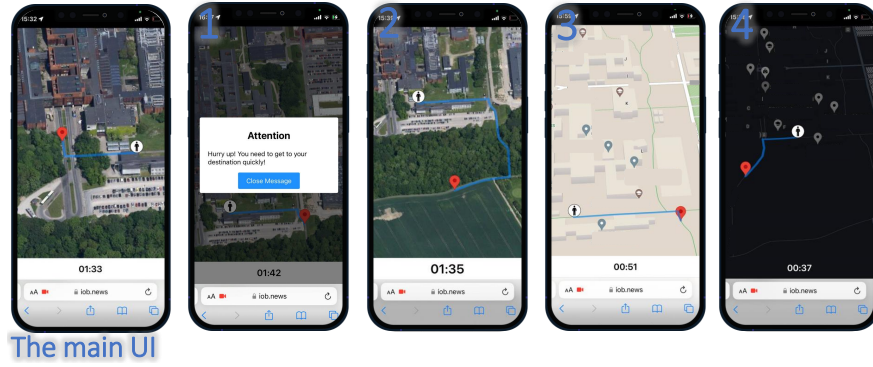


Figure 4: The main UI and the four types of adaptation: 1. Pop-up information, 2. Font Size, 3. Street View, 4. Background color.

emotions, namely joy, fear, and surprise, necessary for emergency training.

Rule 1. (Priority 1): *If (Surprise < disgust) or (Surprise < sad) → Add pop-up information*

Rule 2. (Priority 2): *If (joy < disgust) or (joy < sad) or (joy < anger) → Font Size*

Rule 3. (Priority 3): *If (fear < 0.08) or (disgust > 0.01) → satellite view to street view*

Rule 4. (Priority 4): *If (sadness > fear) → Background color*

5.0.3 Model-Free Reinforcement Learning-based Engine. The MFRL engine implements Algorithm 1 described in Section 3 with parameters $\gamma = 0.9$, $\alpha = 0.5$. These values were fixed based on the sample experiments' overall requirements and feedback. The integer reward/penalty [1, 45] values are assigned as follows: Joy:+5, Fear:+4, Surprise:+3, Neutral:0, Sad:-1, Anger:-2, Disgust:-3. The value assigned is further leveraged to populate the Q-table as depicted in Algorithm 1. We make use of a multi-dimensional array for Q-table. This allows us to select the best possible action from any given state (represented by the combination of current user emotion and running UI configuration). Each possible action, such as changing font size, changing color theme, changing dialog box visibility, etc., is stored in a one-dimensional array. Based on the selected action, the UI is dynamically updated.

6 EXPERIMENTATION

We implemented the app and performed two iterations of experiments to demonstrate the validation of our approach. The survey result (presented in Subsection 6.1) supported the system design and development. The first experiment did not include the task completion rate. We mainly answered two research questions.

- **RQ1.** What UI parameters are associated with specific emotions?
- **RQ2.** How effective is MFRL in implementation (vs. rule-based, sequential, no adaptation)?

RQ2.1. Are the target emotions aroused by different adaptation methods?

RQ2.2. What is the task completion rate associated with each adaptation method?

RQ2.3. What is the correlation between target emotions and task completion?

6.1 Survey Analysis

To ground our system design and development, we designed and deployed an online requirement analysis survey. The two primary aims of conducting the survey were to i) obtain insights on the impact of adaptations on emotions and ii) extract a first Q-Table for the MFRL adaptation engine.

6.1.1 Insights on Emotions and Adaptations. In summary, we aimed to answer the following three questions.

1) What type of information would the participants like to see in the application for learning evacuation?

34% of the participants mentioned that the essential information to show is the evacuation routing, time constraints, and safe evacuation guidelines. 9.6% of the participants found emotional intensity a vital factor for successful evacuation training. These results helped us decide that the main focus should be on evacuation routing based on emotions.

2) What emotions do participants experience during the learning sessions?

46% of participants have already participated in emergency preparedness sessions (including information-sharing sessions and emergency tests). Among the participants with prior learning session experience, 16% found the experience extremely useful. The reason behind the dissatisfaction among other participants was mentioned as a lack of realism of the session and not emotionally engaging sessions (77%). They found the information provided in a vast scope; therefore, the information was forgotten by the sessions participant. Participants reported that they experienced specific emotions in a successful preparedness session: feeling surprised, joy, and fear. The particular report was in line with prior research on the type of emotions involved in emergencies [37]. These results were helpful for us to set target emotions in our adaptation mechanisms. We designed the app in an engaging and realistic way as well.

3) What emotions are involved in various types of adaptation in evacuation learning?

To define the proper type of UI adaptation to emotions, we showed different UI adaptation possibilities to the participants. We showed the specific user interface before adaptation and after adaptation. Next, we asked participants to declare their emotions regarding a specific adaptation. The emotions are based on Ekman's

six basic emotions model on a 7-point Likert scale. The first adaptation concerns changing the font size aroused respectively neutral ($\bar{x}=4.19$), fear ($\bar{x}=3.36$), and surprise ($\bar{x}=3.25$). The second type of adaptation is adding pop-up information for the learner during evacuation. Participants declared feeling surprise ($\bar{x}=4.69$), anger ($\bar{x}=4.03$), and fear ($\bar{x}=3.97$). The third type of adaptation concerns changes in the UI background color. Participants declared neutral ($\bar{x}=4.05$), surprise ($\bar{x}=3.22$), and joy ($\bar{x}=3.08$). The fourth type of adaptation is changing the map from a street to a satellite view. The participant declared anger ($\bar{x}=4.06$), surprise ($\bar{x}=4.03$), and fear ($\bar{x}=3.78$). The final adaptation was the map rerouting, and the results revealed various emotions such as surprise ($\bar{x}=4.0$), neutral ($\bar{x}=3.72$), and fear ($\bar{x}=3.39$). The results of declared emotions engrave the groundwork of the experienced emotions in various adaptations.

6.1.2 Extracting a First Q-Table from the Survey. The downside to a Q-learning approach is that it takes some time for the agent to explore all possible actions from all possible states. In the context of our experiment, since we are working with humans, we cannot simply perform 10000 iterations as training, where we put users through all possible states and actions. Thus, the survey could help extract some general rules and constraints. We implemented constraints into the agent by adding values to the Q-table. The agent will use these Q-values to pick one action over other actions from specific states. For instance, assume our agent has taken action that changed the application's background color to white. In this case, it would almost certainly result in negative emotions from the user if the following action by the agent was to change the text color to white. The reason is that you cannot read white text on a white background. Because of this can be seen as an edge case, a particular state-action pair that we do *not* want the agent to explore. In practice, we would give the action of changing the text color to white a negative Q-value for all where the background state is white. This means the agent will never pick the action of changing the text color to white whenever the background color is white. Apart from this obvious example, our target emotions are joy, surprise, and fear. Based on the survey results, we understood the actions that (based on participants' declaration) could positively and negatively impact those target emotions. So, we set the constraints in a way to facilitate these.

6.2 The Evacuation Experiment's Participants

In the first experiment, twenty-four participants (11 female, 13 male) between 19 to 42 were recruited. In the second experiment, twenty-seven participants (12 female, 15 male) between 20 to 58 were recruited. All participants had prior knowledge of using mobile applications. The experiment took place on individuals' cell phones. Participants held their phones in their hands and checked them continuously while their facial reactions were captured during their physical movement. Participants were provided a link to access EvacuationApp. The experiment started by briefly introducing the use case and explaining the task. Following this section, they were asked to consent to capture their facial expressions and location. The experiment was divided into four phases. Each phase consisted of a display of the map and the route for evacuation. Participants had to follow the route on their mobile screens to complete the evacuation task. Each phase showed a route for evacuation and lasted two minutes maximum (if the task was not completed before).

At the end of each phase, the participant had to answer a short questionnaire. We performed within subject design to compare the performance of our different approaches. Each phase was developed based on a different UI adaptation method, i.e., sequential, MFRL 1, rule-based, and MFRL 2 for *experiment 1*; and MFRL 1, No adaptation, MFRL2, Sequential, MFRL3, and rule-based for *experiment 2*.

6.3 Answer to RQ1: UI Parameters and Emotions

We first computed the average emotion intensity aroused by each type of change in UI parameters to analyze the validity of our defined adaptation with our target emotions. For this RQ, the data is obviously assessed independently of the adaptation method. Table 1/left shows the results of the first experiment, and Table 1/right demonstrates the results of the second experiment. In line with our questionnaire, both experiments confirm that joy intensity was the highest triggered by changes in font size. Fear was mainly triggered by theme color change rather. On average, learners were the most surprised by using the changes in the satellite view. The second type of adaptation that aroused surprise was the pop-up dialogue which was in line with our literature analysis. *The results confirm the hypotheses made in Section 4: the survey research methodology.*

6.4 Answer to RQ2: MFRL v.s. Other Methods

This research question aims at understanding the effectiveness of the MFRL compared with other adaptation methods. For RQ2.1, we have the results of both experiments, but since we integrated the task completion measurement function only for the second experiment, the results of RQ 2.2 and RQ 2.3 will correspond to experiment 2 only.

6.4.1 RQ 2.1: Target Emotions and Adaptation Method. To answer the question of MFRL effectiveness in comparison to other approaches in terms of arousing target emotions, we performed a one-way ANOVA test with the target emotions (surprise, joy, fear) as the dependent variable and the type of experiment (Sequential, RL1, rule-based, RL2 for *experiment 1*; and RL1, no adaptation, RL2, sequential, RL3, rule-based for *experiment 2*) as the independent variable.

Results of Experiment 1. Type of adaptation had a statistically significant effect on surprise as the target emotions $F(3, 2894) = 10.32, p < 0.01$. It also showed a highly significant effect of type of experiment on fear as the target emotion $F(3, 2894) = 9.626, p < 0.01$. The effect of adaptation type on joy as the target emotion was not statistically significant (see the possible reason in the lessons learned Section's discussion). In the next step, we performed a Post-Hoc test using Tukey HSD (Table 2). The result indicated the second iteration of reinforcement learning was significantly more effective in arousing the target emotions (surprise, fear).

Results of Experiment 2. Results indicate that type of experiment had statistically significant effect on all the target emotions with significance level less than 0.05 (Surprise $F(5, 172166) = 2006.3, p < 0.05$, joy $F(5, 172166) = 1919.23, p < 0.05$, Fear $F(5, 172166) = 2086.18, p < 0.05$). To gain a better insight into the significance level of the different adaptation approaches, we performed a Post-Hoc Tukey HSD. The result shown in Table 3 indicated the third iteration of reinforcement learning was significantly more effective

Table 1: Emotion arousal and the changes in UI.

Descriptive Statistics (Experiment 1)					Descriptive Statistics (Experiment 2)				
	Adaptation Name	Mean	Std. Deviation	N		Adaptation Name	Mean	Std. Deviation	N
Surprise	ColorThemeDark	0.001654973	0.00210981	22	Surprise	ColorThemeDark	0.000239375	0.00041136	23
	FontSizeLarge	0.003478613	0.00569257	22		FontSizeLarge	0.00018913	0.00036682	22
	MapTypeSatellite	0.005159434	0.01357806	23		MapTypeSatellite	0.00439321	0.01178233	22
	PopupVisible	0.003611541	0.00901929	24		PopupVisible	0.000522933	0.0013886	24
	Total	0.003497614	0.00871749	91		Total	0.001306236	0.00600388	91
Joy	ColorThemeDark	0.121724833	0.19302968	22	Joy	ColorThemeDark	0.17780563	0.38053943	23
	FontSizeLarge	0.139943573	0.23440106	22		FontSizeLarge	0.072975863	0.21773765	22
	MapTypeSatellite	0.114505343	0.18985003	23		MapTypeSatellite	0.099668837	0.27790235	22
	PopupVisible	0.115723079	0.18370163	24		PopupVisible	0.011411343	0.02484915	24
	Total	0.122721777	0.19761571	91		Total	0.089687748	0.26143691	91
fear	ColorThemeDark	0.004183614	0.01196534	22	fear	ColorThemeDark	0.000122874	0.0002509	23
	FontSizeLarge	0.001065816	0.00318769	22		FontSizeLarge	0.001794525	0.00733253	22
	MapTypeSatellite	0.001279343	0.00379304	23		MapTypeSatellite	0.000247322	0.00109847	22
	PopupVisible	0.001226463	0.00471166	24		PopupVisible	4.67694E-05	0.00015545	24
	Total	0.001915906	0.00682871	91		Total	0.000537024	0.00365561	91

Table 2: Post-Hoc analysis for target emotions and adaptation method in the first experiment.

Multiple Comparisons					
Tukey HSD					
Dependent Variable	Experiment (I)	Experiment (J)	Mean Difference (I-J)	Std. Error	Sig.
Surprise	RL2nd	Sequential	.004529278438771*	0.000834703	<0.01
		RL1st	.003489306356198*	0.000871724	<0.01
		Rule-based	.002495088155604*	0.000905264	0.03
		Sequential	.001269085063540*	0.000242153	<0.01
Fear	RL2nd	RL1st	.000992190093620*	0.000252893	<0.01
		Rule-based	.000727134669543*	0.000262623	0.029
		Sequential	.001269085063540*	0.000242153	<0.01

*. The mean difference is significant at the 0.05 level.

in arousing the target emotions. Moreover, the results indicated that the third iteration of the MFRL approach is performing better than the first and second approaches. This result aligns with our expectation that the MFRL approach refines itself in each iteration.

6.4.2 RQ 2.2: Task Completion and Adaptation Method. Our second factor to examine the effectiveness of the MFRL is the completion rate in types of adaptation approaches. To this end, we performed a one-way ANOVA test, having the completion rate as a dependent variable and the type of experiment (RL1, no adaptation, RL2, sequential, RL3, rule-based) as an independent variable. Our result showed that the experiment type had a statistically significant effect on the task completion rate $F(5, 166) = 2711, p < 0.05$. In the next step, we are particularly interested to see if the difference between groups of adaptation types is significant or not. To this end, we performed a Post-Hoc test to examine the last MFRL iteration with the other adaptation approaches and the previous MFRL iterations (See Table 4). The between-group analysis revealed a significant difference between the third MFRL iteration and the other types of adaptations. It showed the third iteration of the MFRL approach outperforms the other adaptation methods. Moreover, the results show that the third iteration of MFRL also improved compared to the first and second MFRL iterations. This is primarily because, in the case of MFRL, continuous learning is happening as opposed to the other adaptation approaches. In the case of a sequential and rule-based adaptation, the adaptation performed is more based on a standard pre-fixed pattern or a choice made depending on pre-configured rules. However, in MFRL, the feedback obtained after each adaptation (through the user emotion captured after every

adaptation cycle) continuously improves the adaptation process, which is further enhanced after each iteration.

6.4.3 RQ 2.3: Target Emotions and Task Completion Correlation. Finally, we were interested in understanding the correlation between emotions and task completion rate. As shown in Table 5, we measured the correlation between the target emotions and the evacuation completion rate. The results show a positive correlation between the target emotions and task completion.

7 LESSONS LEARNED

Putting humans and their emotions as a primary concern of adaptive systems enhances their quality of experience. In the process of developing the adaptation approach and then experimenting using the case study, we learned that:

- Humans have different goals, desires, characteristics, and previous experiences. Thus, creating a general model for human behaviors is difficult. In addition, individuals may show various behaviors to a single stimulus if they are in different physical or social contexts. In our experiment, we understood that the MFRL agent improves in iterations by learning and ameliorates the decision for individuals. This was further demonstrated by the results presented in RQ2. When using our adaptation engine for your UI adaptation, let the MFRL learn in several iterations to provide you with the best results.
- Capturing emotions from only the face may cause an accuracy challenge. The issue becomes more severe when emotions are captured with mobile phones' built-in cameras, in movement, and while evacuating. In fact, people do not always look at their phones when they complete a task. However, equipping people

Table 3: Post-Hoc analysis for target emotions and adaptation method in the second experiment.

Multiple Comparisons						
Dependent Variable		Mean Difference (I-J)		Std. Error	Sig.	95% Confidence Interval Lower Bound Upper Bound
Surprise	Tukey HSD 3rd RL iteration	1st RL iteration	.006204148230151*	0.0000914911	<0.5	0.0059434 0.0064649
		No adaptation	.007571266605247*	0.0000914911	<0.5	0.0073105 0.0078320
		2nd RL iteration	.003882242690194*	0.0000931700	<0.5	0.0036167 0.0041478
		Sequential adaptation	.006804342119558*	0.0000914911	<0.5	0.0065436 0.0070651
		Rule-based adaptation	.007245862295789*	0.0000914911	<0.5	0.0069851 0.0075066
Joy	Tukey HSD 3rd RL iteration	1st RL iteration	.061253131394917*	0.0021700618	<0.5	0.0550690 0.0674372
		No adaptation	.192931651526901*	0.0021700618	<0.5	0.1867475 0.1991158
		2nd RL iteration	.058729068746643*	0.0022098828	<0.5	0.0524315 0.0650267
		Sequential adaptation	.023513915947909*	0.0021700618	<0.5	0.0173298 0.0296980
		Rule-based adaptation	.085097473183649*	0.0021700618	<0.5	0.0789134 0.0912816
Fear	Tukey HSD 3rd RL iteration	1st RL iteration	.004539045245759*	0.0000565092	<0.5	0.0043780 0.0047001
		No adaptation	.002276451255905*	0.0000565092	<0.5	0.0021154 0.0024375
		2nd RL iteration	.004541049793804*	0.0000575462	<0.5	0.0043771 0.0047050
		Sequential adaptation	.004569287272770*	0.0000565092	<0.5	0.0044083 0.0047303
		Rule-based adaptation	.003686157440484*	0.0000565092	<0.5	0.0035251 0.0038472

*. The mean difference is significant at the 0.05 level.

Table 4: Post-Hoc analysis for task completion rate.

Multiple Comparisons					
Dependent Variable	Experiment (I)	Experiment (J)	Mean Difference (I-J)	Std. Error	Sig.
Completion rate	RL3rd	RL1st	15.11139*	4.142338882	<0.01
		No adaptation	6.85037037	3.249517968	<0.05
		RL2nd	12.69889*	4.142338882	<0.05
		Sequential	8.464444*	3.249517968	<0.05
		Rule-based	8.22037*	3.293051376	<0.05

*. The mean difference is significant at the 0.05 level.

Table 5: Correlation analysis between emotions and task completion rate.

Correlations					
Variable	Correlation	Count	Statistic		
			Lower C.I.	Upper C.I.	
Completion rate	Surprise	0.149	27	-0.245	0.500
	Joy	0.131	27	-0.262	0.487
	Fear	0.109	27	-0.282	0.470

Missing value handling: PAIRWISE, EXCLUDE. C.I. Level: 95.0

with additional sensors would be uncomfortable, especially when people move. Therefore, if your application works fine on a desktop, using multimodal emotion detection approaches to feed our MFRL engine can help achieve better results. In our future work, we investigate how the Internet of Things devices could help achieve better emotional accuracy.

- Privacy is a significant challenge in systems that gather human data. At the beginning of our research, when we conducted pilot experiments with our colleagues, we started by uploading the videos to be processed on a server. This raised concerns regarding privacy and performance. In the following steps (including the two iterations reported), we modified our architecture to capture and process facial expressions locally (on devices) and transferred only the emotions and their intensity. We also assign IDs to avoid any possibility of identifying individuals.
- Along with automatic detection, collecting insights from users by self-reporting methods gives a better picture of the system under development. We gathered users' feedback after each experiment and in each iteration of experiments and improved our design, development, and experiments process. We also created a first Q-Table for our MFRL based on declarative emotions associated with UIs. This enhanced the performance of our adaptation approach. This was demonstrated in the results in Table 2 and Table 3 where we can clearly see how having an initial Q-table enhanced the adaptation effectiveness. Thus, depending on the nature of your

case, we suggest adopting an iterative approach with humans in the loop and enhancing your interaction and UI in iterations.

- In the experiments, we noticed that people get frustrated with too many adaptations, which may negatively impact their mood and performance. For instance, adopting a rule-based approach that does not consider temporal aspects to, e.g., set the maximum number of adaptations does not really work in practice. Using our MFRL solves this problem, but if you still use the traditional adaptation approaches, consider the temporal aspects of adaptation.

8 THREATS TO VALIDITY

Threats to construct validity concern using controlled experiments and incorrect adaptations that may arise from using an MFRL technique. Although we have performed a controlled experiment, we have tried to emulate a realistic scenario regarding tasks provided to users, time taken, and the path to be followed. To address the problem of incorrect adaptations, the Q-table was populated with initial values based on feedback obtained from sample experiments. **Threats to internal validity** concern using only basic emotions and using a library like FaceAPI, which may have accuracy issues with emotion detection, particularly with different types of populations due to bias in the training set. During our sample experiment, we did ensure that such bias did not exist. Further study is required to measure the impact of such issues on the overall adaptation effectiveness. For instance, using multimodal emotion detection approaches to feed our MFRL engine can help achieve better results. Moreover, depending on the population, finer tweaks can be made. **Threats to external validity** concern the generalizability and scalability of our approach. Although our approach has been evaluated on a well-defined experiment, the approach can be generalized to different scenarios as long as the application provides means to capture user emotions and mechanisms to adapt the user interface. Further experiments are required to evaluate our approach to different scenarios and to test the effectiveness

and efficiency issues that may arise from increased state and action space.

9 CONCLUSION

This paper presented an emotional UI adaptation approach based on model-free reinforcement learning. The approach is conceptualized based on the MAPE-K control loop concept, where an adaptation supervisor interacts with the users and their devices to change the UIs and improve the users' feelings and performance. We compared the MFRL with traditional adaptation approaches such as sequential and rule-based. We applied the approach in a case study where the users used a mobile application that trains users on safe evacuation from a danger zone. The results demonstrated that various UIs could cause specific emotions in users. In addition, we proved that the MFRL-based adaptation approach functions better in completing evacuation tasks and arousing target emotions. In future work, we will apply our adaptation engine to more domains, such as human-robot interaction.

Acknowledgements

This work is supported by the Innovation Fund Denmark for the project DIREC (9142-00001B).

REFERENCES

- [1] Mina Alipour, Sophie Dupuy-Chessa, and Eric Céret. 2021. An emotion-oriented problem space for ui adaptation: From a literature review to a conceptual framework. In *2021 9th International Conference on Affective Computing and Intelligent Interaction (ACII)*. IEEE, 1–8.
- [2] Mina Alipour, Sophie Dupuy-Chessa, and Eline Jongmans. 2020. Disaster mitigation using interface adaptation to emotions: a targeted literature review. In *10th International Conference on the Internet of Things Companion*. 1–15.
- [3] Reem Alnanih, Olga Ormandjieva, and Thiruvengadam Radhakrishnan. 2013. Context-based and rule-based adaptation of mobile user interfaces in mHealth. *Procedia Computer Science* 21 (2013), 390–397.
- [4] Claudio Arbib, Davide Arcelli, Julie Dugdale, Mahyar T Moghaddam, and Henry Muccini. 2019. Real-time emergency response through performant IoT architectures. In *International Conference on Information Systems for Crisis Response and Management (ISCRAM)*.
- [5] Claudio Arbib, Mahyar T Moghaddam, and Henry Muccini. 2019. Iot flows: a network flow model application to building evacuation. *A View of Operations Research Applications in Italy, 2018* (2019), 115–131.
- [6] Claudio Arbib, Henry Muccini, and Mahyar Tourchi Moghaddam. 2018. Applying a network flow model to quick and safe evacuation of people from a building: a real case. *RSFF* 18 (2018), 50–61.
- [7] Jonathan S Barnhoorn, Erwin Haasnoot, Bruno R Bocanegra, and Henk van Steenbergen. 2015. QRTengine: An easy solution for running online reaction time experiments using Qualtrics. *Behavior research methods* 47, 4 (2015), 918–929.
- [8] Margaret M Bradley and Peter J Lang. 1994. Measuring emotion: the self-assessment manikin and the semantic differential. *Journal of behavior therapy and experimental psychiatry* 25, 1 (1994), 49–59.
- [9] Tobias Brosch, David Sander, Gilles Pourtois, and Klaus R Scherer. 2008. Beyond fear: Rapid spatial orienting toward positive emotional stimuli. *Psychological science* 19, 4 (2008), 362–370.
- [10] Hartmut Dieterich, Uwe Malinowski, Thomas Kühme, and Matthias Schneider-Hufschmidt. 1993. State of the art in adaptive user interfaces. *Human factors in information technology* 10 (1993), 13–13.
- [11] Julie Dugdale, Mahyar T Moghaddam, and Henry Muccini. 2020. Human behaviour centered design: developing a software system for cultural heritage. In *Proceedings of the ACM/IEEE 42nd International Conference on Software Engineering: Software Engineering in Society*. 85–94.
- [12] Paul Ekman. 1993. Facial expression and emotion. *American psychologist* 48, 4 (1993), 384.
- [13] Paul Ekman. 1999. Basic emotions. *Handbook of cognition and emotion* 98, 45–60 (1999), 16.
- [14] Paul Ekman, Wallace V Friesen, Maureen O'sullivan, Anthony Chan, Irene Diacoyanni-Tarlatzis, Karl Heider, Rainer Krause, William Ayhan LeCompte, Tom Pitcairn, Pio E Ricci-Bitti, et al. 1987. Universals and cultural differences in the judgments of facial expressions of emotion. *Journal of personality and social psychology* 53, 4 (1987), 712.
- [15] Sergio Firmenich, Alejandra Garrido, Fabio Paternò, and Gustavo Rossi. 2019. User interface adaptation for accessibility. In *Web Accessibility*. Springer, 547–568.
- [16] Julián Andrés Galindo, Sophie Dupuy-Chessa, Nadine Mandran, and Eric Céret. 2018. Using user emotions to trigger UI adaptation. In *2018 12th International Conference on Research Challenges in Information Science (RCIS)*. IEEE, 1–11.
- [17] Jose Maria Garcia-Garcia, Victor MR Penichet, and Maria D Lozano. 2017. Emotion detection: a technology review. In *Proceedings of the XVIII international conference on human computer interaction*. 1–8.
- [18] Jan Gläscher, Nathaniel Daw, Peter Dayan, and John P O'Doherty. 2010. States versus rewards: dissociable neural prediction error signals underlying model-based and model-free reinforcement learning. *Neuron* 66, 4 (2010), 585–595.
- [19] Donald Glowinski, Antonio Camurri, Gualtiero Volpe, Nele Dael, and Klaus Scherer. 2008. Technique for automatic emotion recognition by body gesture analysis. In *2008 IEEE Computer society conference on computer vision and pattern recognition workshops*. IEEE, 1–6.
- [20] Brad Hardin and Dave McCool. 2015. *BIM and construction management: proven tools, methods, and workflows*. John Wiley & Sons.
- [21] Sabina Hunziker, Laura Laschinger, Simone Portmann-Schwarz, Norbert K Semmer, Franziska Tschan, and Stephan Marsch. 2011. Perceived stress and team performance during a simulated resuscitation. *Intensive care medicine* 37, 9 (2011), 1473–1479.
- [22] Jeffrey O Kephart and David M Chess. 2003. The vision of autonomic computing. *Computer* 36, 1 (2003), 41–50.
- [23] Yanal Mahmoud Kilani. 2021. SMART Business Role in Supporting Marketing Strategies among Telecommunication Organizations In Jordan. *Turkish Journal of Computer and Mathematics Education (TURCOMAT)* 12, 6 (2021), 4568–4582.
- [24] Paul R Kleinginna and Anne M Kleinginna. 1981. A categorized list of emotion definitions, with suggestions for a consensual definition. *Motivation and emotion* 5, 4 (1981), 345–379.
- [25] Amit Konar and Aruna Chakraborty. 2015. *Emotion recognition: A pattern analysis approach*. John Wiley & Sons.
- [26] Jonathan Lazar, Jinjuan Heidi Feng, and Harry Hochheiser. 2017. *Research methods in human-computer interaction*. Morgan Kaufmann.
- [27] Baihan Lin, Guillermo Cecchi, Djallel Bouneffouf, Jenna Reinen, and Irina Rish. 2019. A story of two streams: Reinforcement learning models from human behavior and neuropsychiatry. *arXiv preprint arXiv:1906.11286* (2019).
- [28] David Lindlbauer, Anna Maria Feit, and Otmarr Hilliges. 2019. Context-aware online adaptation of mixed reality interfaces. In *Proceedings of the 32nd annual ACM symposium on user interface software and technology*. 147–160.
- [29] Tiago V Maia and Michael J Frank. 2011. From reinforcement learning models to psychiatric and neurological disorders. *Nature neuroscience* 14, 2 (2011), 154–162.
- [30] Christian Martin, Sanim Rashid, and Christian Herdin. 2016. Designing responsive interactive applications by emotion-tracking and pattern-based dynamic user interface adaptation. In *International Conference on Human-Computer Interaction*. Springer, 28–36.
- [31] Elizabeth Martin. 2006. Survey questionnaire construction. *Survey methodology* 13 (2006), 2006.
- [32] Sascha Meudt, Miriam Schmidt-Wack, Frank Honold, Felix Schüssel, Michael Weber, Friedhelm Schwenker, and Günther Palm. 2016. Going further in affective computing: how emotion recognition can improve adaptive user interaction. In *Toward Robotic Socially Believable Behaving Systems-Volume I*. Springer, 73–103.
- [33] Nesrine Mezhoui. 2013. User interface adaptation based on user feedback and machine learning. In *Proceedings of the companion publication of the 2013 international conference on intelligent user interfaces companion*. 25–28.
- [34] Mahyar T Moghaddam, Mina Alipour, and Mikkel Baun Kjærgaard. 2023. User Interface and Architecture Adaption Based on Emotions and Behaviors. In *2023 IEEE 20th International Conference on Software Architecture Companion (ICSA-C)*. IEEE, 101–105.
- [35] Mahyar T Moghaddam, Henry Muccini, and Julie Dugdale. 2022. Intelligent Building Evacuation: From Modeling Systems to Behaviors. In *Disaster Management and Information Technology: Professional Response and Recovery Management in the Age of Disasters*. Springer, 111–129.
- [36] Mahyar T Moghaddam, Henry Muccini, Julie Dugdale, and Mikkel Baun Kjærgaard. 2022. Designing Internet of Behaviors Systems. In *2022 IEEE 19th International Conference on Software Architecture (ICSA)*. IEEE, 124–134.
- [37] Patricia Moravec, Lu Lucy Yan, and Alfonso Pedraza Martinez. 2021. Wildfire Response Operations: Intentional Fear Reduction through Social Media Updates. *Kelley School of Business Research Paper* 2021-05 (2021).
- [38] Vivian Genaro Motti and Jean Vanderdonckt. 2013. A computational framework for context-aware adaptation of user interfaces. In *IEEE 7th International Conference on Research Challenges in Information Science (RCIS)*. IEEE, 1–12.
- [39] Henry Muccini, Romina Spalazzese, Mahyar T Moghaddam, and Mohammad Sharaf. 2018. Self-adaptive IoT architectures: An emergency handling case study. In *Proceedings of the 12th European Conference on Software Architecture: Companion Proceedings*. 1–6.

- [40] Daniel M Oppenheimer, Tom Meyvis, and Nicolas Davidenko. 2009. Instructional manipulation checks: Detecting satisficing to increase statistical power. *Journal of experimental social psychology* 45, 4 (2009), 867–872.
- [41] Jonathan Posner, James A Russell, and Bradley S Peterson. 2005. The circumplex model of affect: An integrative approach to affective neuroscience, cognitive development, and psychopathology. *Development and psychopathology* 17, 3 (2005), 715–734.
- [42] Roger Pressman and Bruce Maxim. 2019. *ISE Software Engineering: A Practitioner's Approach* (9 ed.). McGraw-Hill Education, Columbus, OH.
- [43] Arman Savran, Koray Ciftci, Guillaume Chanel, Javier Mota, Luong Hong Viet, Blent Sankur, Lale Akarun, Alice Caplier, and Michele Rombaut. 2006. Emotion detection in the loop from brain signals and facial images. (2006).
- [44] Annett Schirmer and Ralph Adolphs. 2017. Emotion perception from face, voice, and touch: comparisons and convergence. *Trends in cognitive sciences* 21, 3 (2017), 216–228.
- [45] Hanan Shteingart and Yonatan Loewenstein. 2014. Reinforcement learning and human behavior. *Current Opinion in Neurobiology* 25 (2014), 93–98.
- [46] Jean-Sébastien Sottet, Gaëlle Calvary, Jean-Marie Favre, Joëlle Coutaz, Alexandre Demeure, and Lionel Balme. 2006. Towards Model Driven Engineering of Plastic User Interfaces. In *Satellite Events at the MoDELS 2005 Conference*, Jean-Michel Bruel (Ed.). Springer Berlin Heidelberg, Berlin, Heidelberg, 191–200.
- [47] Jean-Sébastien Sottet, Vincent Ganneau, Gaëlle Calvary, Joëlle Coutaz, Alexandre Demeure, Jean-Marie Favre, and Rachel Demumieux. 2007. Model-driven adaptation for plastic user interfaces. In *IFIP Conference on Human-Computer Interaction*. Springer, 397–410.
- [48] Hamed Taherdoost. 2016. How to design and create an effective survey/questionnaire; A step by step guide. *International Journal of Academic Research in Management (IJARM)* 5, 4 (2016), 37–41.
- [49] Kashyap Todi, Gilles Bailly, Luis Leiva, and Antti Oulasvirta. 2021. Adapting user interfaces with model-based reinforcement learning. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [50] Christopher JCH Watkins and Peter Dayan. 1992. Q-learning. *Machine learning* 8, 3-4 (1992), 279–292.
- [51] Danny Weyns, Bradley Schmerl, Vincenzo Grassi, Sam Malek, Raffaella Mirandola, Christian Prehofer, Jochen Wuttke, Jesper Andersson, Holger Giese, and Karl M Göschka. 2013. On patterns for decentralized control in self-adaptive systems. In *Software Engineering for Self-Adaptive Systems II*. Springer, 76–107.
- [52] Timothy D Wilson and Daniel T Gilbert. 2003. Affective forecasting. (2003).
- [53] Eldad Yechiam, Jerome R Busemeyer, Julie C Stout, and Antoine Bechara. 2005. Using cognitive models to map relations between neuropsychological disorders and human decision-making deficits. *Psychological science* 16, 12 (2005), 973–978.
- [54] Enes Yigitbas, André Hottung, Sebastian Mansfield Rojas, Anthony Anjorin, Stefan Sauer, and Gregor Engels. 2019. Context-and data-driven satisfaction analysis of user interface adaptations based on instant user feedback. *Proceedings of the ACM on Human-Computer Interaction* 3, EICS (2019), 1–20.
- [55] Mao Zheng, Qian Xu, and Hao Fan. 2016. Modeling The Adaption Rule in Context-aware Systems. *arXiv preprint arXiv:1609.01614* (2016).